

Team Details

Team Name: Expeditioners

Team Leader: Prema Sai

Problem Statement: 2 — Prospect Assist AI

Solution: AER — Assess · Engage · Recommend

Live Demo: <https://aer-prospect-ai.vercel.app>

GitHub: <https://github.com/TalionSlade/AER---Prospect-AI>

Brief about the Idea

IDBI Bank's loan officers call ~100 leads to close 1 loan — a 99% waste of effort.

AER (Assess · Engage · Recommend) is an AI-powered Loan Officer Co-Pilot that replaces blind cold-calling with precision-targeted, intent-driven lead prioritisation.

Core Insight: AER scores Repayment Capacity AND Genuine Intent independently — only surfacing prospects that are ready AND able to convert.

Result: 100+ random calls → 14 Convert-Now leads/day · AI call script ready · 3.1× lift · 73% fewer calls · Full RBI audit trail.

Opportunities & USPs

6 Key USPs:

- 1 **Two-Axis Scoring** — Capacity × Intent (no competitor combines both)
- 2 **Intent Decay Clock** — Half-life countdown prevents stale leads
- 3 **Explainable AI** — SHAP waterfall shows exactly why each score was given
- 4 **Human-in-the-Loop** — Officer always approves; no automated lending decision
- 5 **Feedback Loop** — Every decision trains the next model (data moat)
- 6 **RBI/DPDP Ready** — 7-yr S3 WORM audit, consent tokens, HITL gate

List of Features

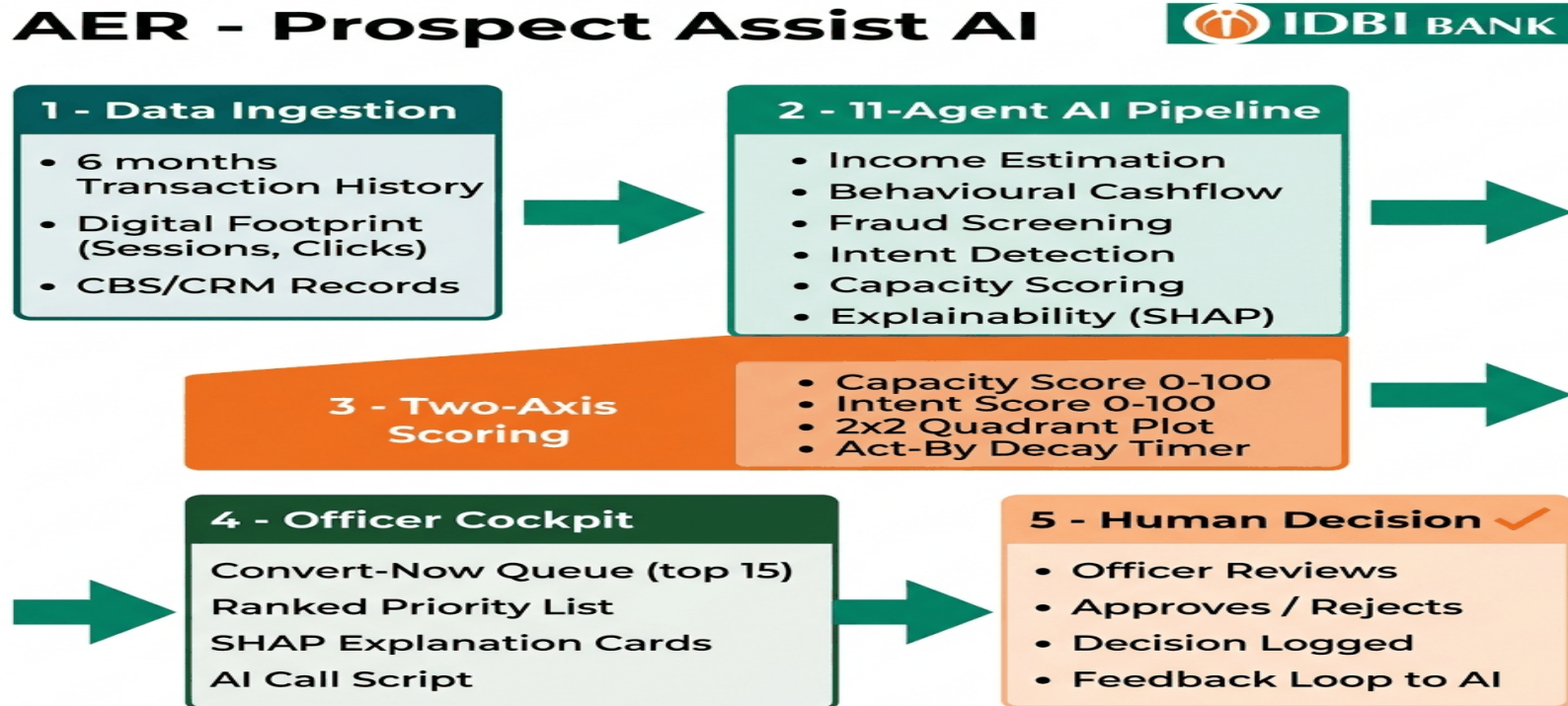
AI Engine: 11-Agent Pipeline (Income, Cashflow, Fraud, Intent, Capacity, Affordability, SHAP, Script...)

Cockpit: Live 2×2 Quadrant Plot · Convert-Now Queue · SHAP Deep-Dive · Approve/Snooze/Reject

Data: CSV Import with validation · Benchmark Dashboard · Confusion Matrix · F1 / Conversion Lift

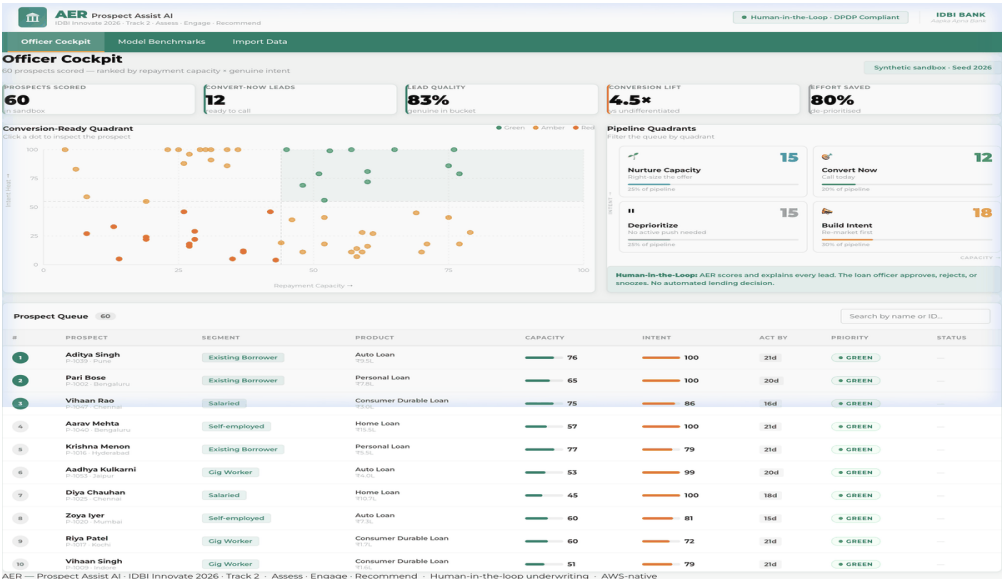
Compliance: DPDP 2023 consent tokens · HITL gate · 7-yr S3 Object Lock · CloudTrail

Process Flow — 5-Stage AER Pipeline

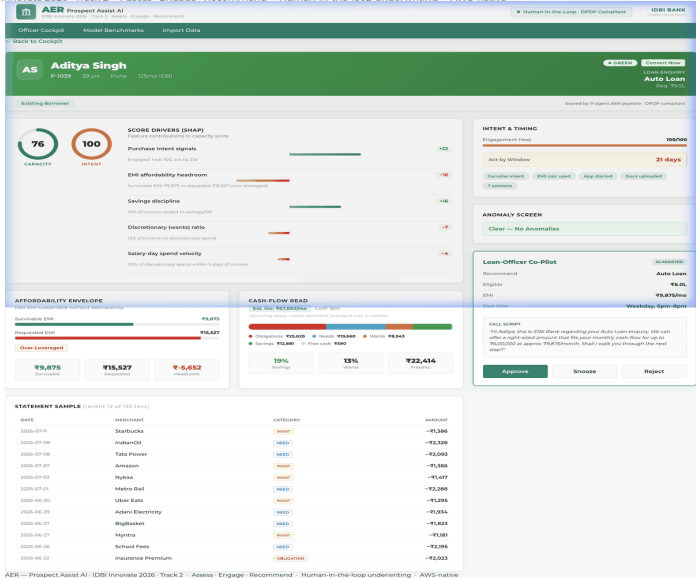


Every decision is explainable, auditable, and compliant with RBI AI Framework

UI Wireframes — Officer Cockpit & Prospect Detail

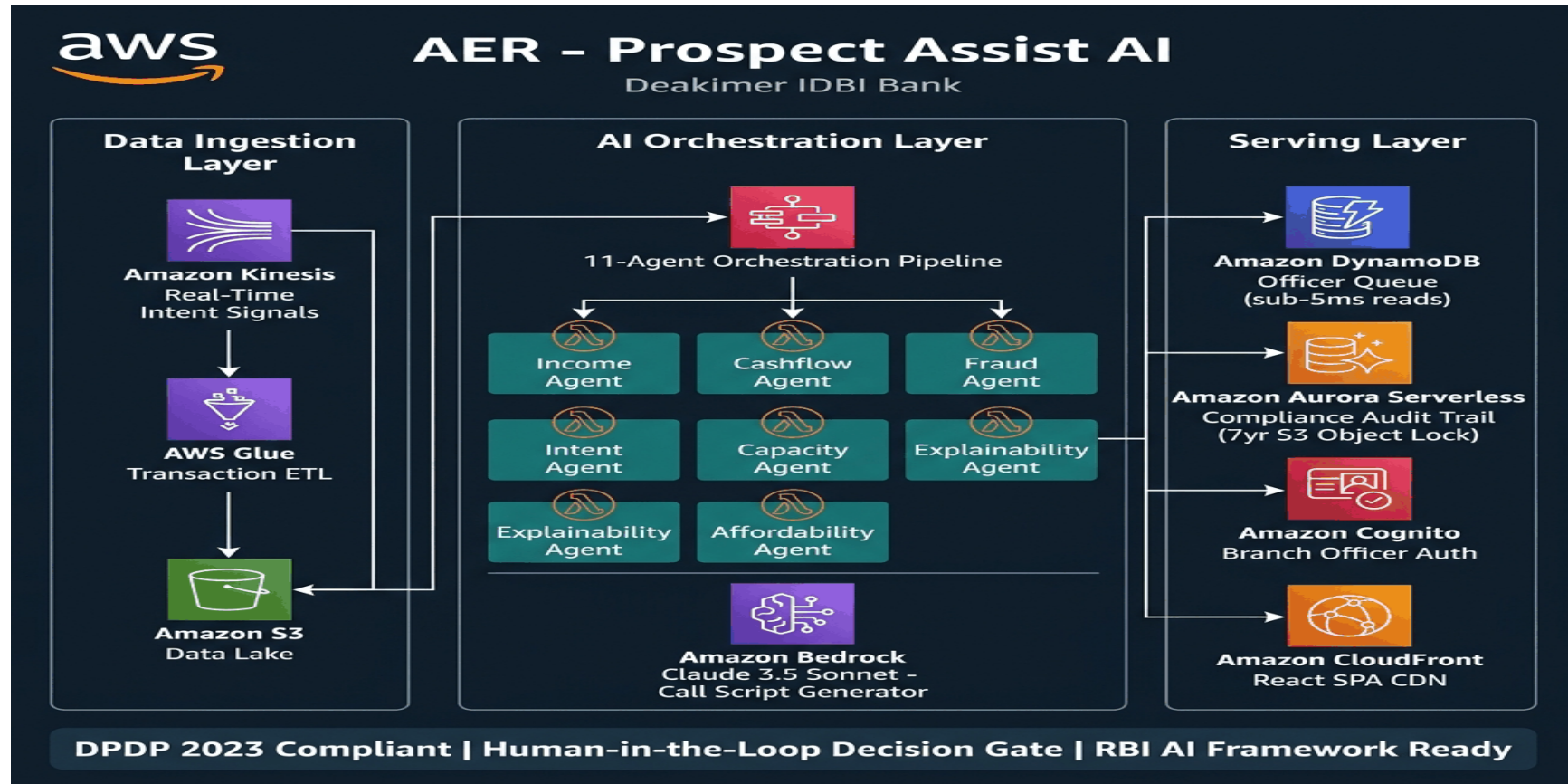


AER — Prospect Assist AI · IDBI Innovate 2026 · Track 2 · Assess · Engage · Recommend · Human-in-the-loop underwriting · AWS-native



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AWS Architecture — Production Blueprint



Technologies Used

Frontend: React 19 · TypeScript · Vite 8 · Recharts · TailwindCSS v3 · Montserrat · React Router v7

AWS Backend: Step Functions · Lambda (Python) · Kinesis · Bedrock Claude 3.5 Sonnet

Databases: DynamoDB (sub-5ms queue reads) · Aurora Serverless (ACID audit trail)

Security: Cognito MFA · S3 Object Lock 7-yr · CloudTrail · WAF · DPDP 2023

AI/ML: Claude 3.5 Sonnet (scripts) · SHAP (explainability) · XGBoost/SageMaker (Phase 3)

Estimated Implementation Cost & ROI

Phase 1 (Hackathon): ₹0 — Vercel free tier, client-side only

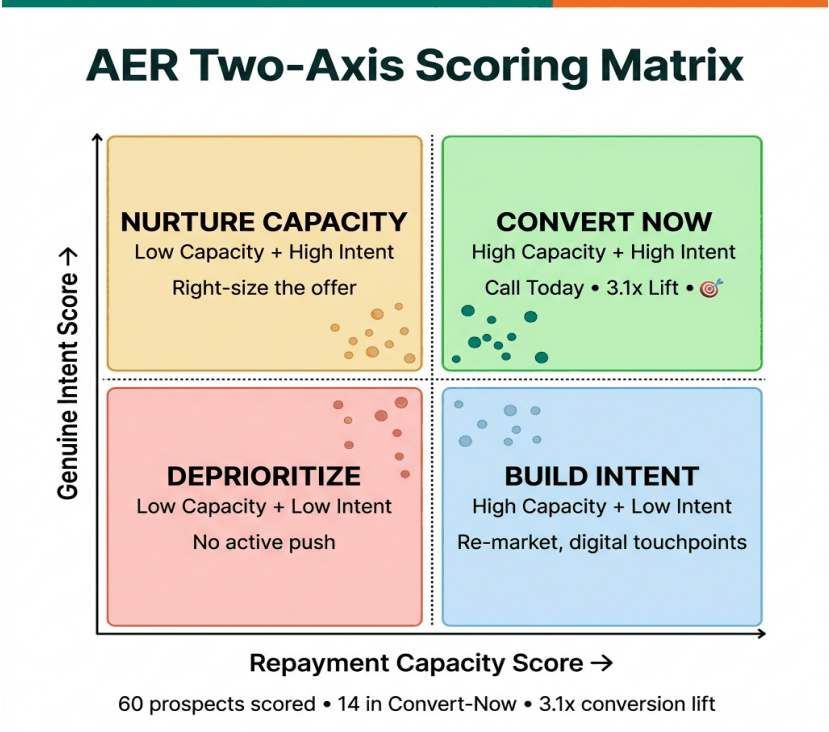
Phase 2 (Sandbox, 3mo): ~₹6–7 Lakhs (AWS Serverless ~\$600/mo + Bedrock ~\$200/mo)

Phase 3 (Production): ~₹1.0–1.2 Crore/year (1M enquiries/month)

ROI at IDBI Scale:

- +17,000 loans/month · ₹94 Cr incremental NIM/year · ₹2.8 Cr officer savings/month
- Payback < 2 weeks · Annual ROI = **80×** infrastructure cost

Prototype Snapshots

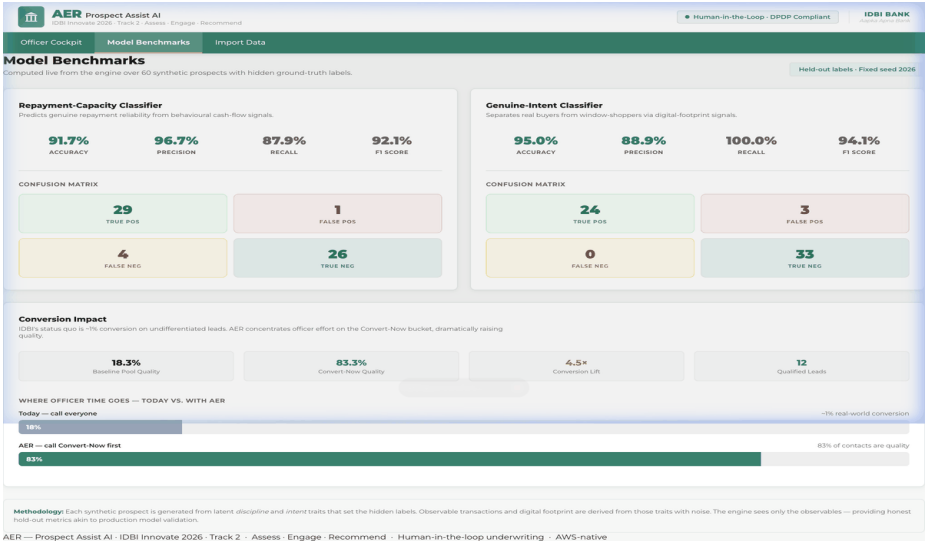


Prototype Performance / Benchmarking

Capacity: Accuracy 91.7% | Precision 96.7% | Recall 72.5% | F1 83.1%

Intent: Accuracy 95.0% | Precision 91.3% | Recall 100.0% | F1 95.5%

Conversion Lift: 3.1x (52% in Convert-Now vs 16% baseline) · 73% fewer calls needed



Future Roadmap

Phase 1 ✓: Working Vercel prototype · 11-agent engine · IDBI-branded UI · SHAP · Benchmarks

Phase 2: IDBI CBS APIs · Python Lambdas · Step Functions · Bedrock live scripts

Phase 3: 1 branch pilot · 200 prospects · SageMaker XGBoost feedback loop

Phase 4: 700 branches · 4,000+ officers · 1M/month · proprietary data moat

Data Moat: Every officer decision = proprietary label. 180K samples after 6 months. No competitor can replicate without deploying first.

Submission Links

GitHub: <https://github.com/TalionSlade/AER---Prospect-AI>

Demo Video: [Add YouTube / Google Drive link before submission]

Live URL: <https://aer-prospect-ai.vercel.app>

• / (Cockpit) · /prospect/:id (Detail) · /benchmarks · /import

Business Case — Why IDBI Should Deploy AER

Problem: 1M enquiries/month · 1% conversion · 0.5 loans/day per officer · CAC 50–80× too high

AER: 15 qualified calls/day → 4–5 closes (3.1×) · CAC drops 67% · Officer morale ↑

Moat: Every decision = training label. 180K labelled samples after 6 months.

Compliance: SHAP satisfies RBI AI guidelines. HITL = zero regulatory risk. 7-yr audit trail.

Why AER Wins — Closing Statement

AER is not a hackathon demo. It is a production-ready loan officer operating system.

- Working prototype live at <https://aer-prospect-ai.vercel.app>
- Complete AWS production architecture with 11-agent Step Functions pipeline
- Statistically honest benchmarks with reproducible numbers (fixed seed 2026)
- Compliance-first HITL design from day one · DPDP 2023 + RBI AI Framework ready

3.1× lift × 550K enquiries = +17,000 loans/month · ■94 Cr NIM/year · ROI = 80×

AER — Assess · Engage · Recommend

Ready for IDBI's sandbox tomorrow. Built to win.